

Designing a Motion-Onset Visual-Evoked Potential-Based Brain–Computer Interface to Control a Computer Game

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Abstract—Brain–computer interface (BCI) establishes an alternative direct communication channel between the brain and computer or other external devices. Motion-onset visual-evoked potential (mVEP) related to the individual perception of movement is among synchronous BCI system that utilizes external stimulus. mVEPs stimulation paradigm is like an oddball stimulation paradigm. Researchers have recently been investigating mVEPs to control a computer game. To control a computer game, they used translation stimulus to evoke these potentials in electroencephalography brain signals. In this study, we examine designing a computer game with four different types of stimuli, including translation, contraction, expansion, and rotation. We also used the time-domain and wavelet coefficients for the feature extraction. Also, we used different types of linear discriminant analysis (LDA) classification methods, including least absolute shrinkage and selection operator, stepwise LDA (SWLDA); moreover, and spatial–temporal discriminant analysis (STDA) for detecting the target stimulus. Our results demonstrate that contraction, expansion, and rotation stimuli, as well as translation, could be employed to control a BCI game. Our results showed that subjects got higher scores, game accuracy control, and information transfer rate using the rotation stimulus with SWLDA/STDA classification methods.

Index Terms—Brain–computer interface (BCI), computer game, electroencephalography (EEG), motion-onset visually evoked potentials (mVEP).

I. INTRODUCTION

BRAIN–computer interfaces (BCIs) create a direct channel between the electrophysiological signals of the brain and computers or other external devices, turning electrophysiological signals of the brain into computer commands [1]. Patients with spinal cord or other nervous system injuries or diseases can use these systems to improve their quality of life [2], [3]. Meanwhile, healthy people can use it for entertainment purposes [4], [5], [6].

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A lot of electrophysiological signals of the brain, such as electroencephalography (EEG) [7], magnetoencephalography [8], functional magnetic resonance imaging [9], functional near-infrared spectroscopy [10], electrocorticography [11], local field potentials, and single-unit activity, have been used in BCI systems, recently. EEG-based BCI systems have valuable advantages due to their low cost and high portability [12].

Visual-evoked potential-based BCI systems are a subset of BCIs built by presenting visual stimuli to evoke potentials in brain signals. A computer monitor can present these stimuli. As a subset of VEP-based BCIs, we can mention P300, steady-state visually evoked potential (SSVEP), and motion-onset visual-evoked potential (mVEP) based BCIs, which have been successful in spelling applications [13], [14] or neurogame control [15], [16], [17], [18]. P300 or SSVEP with flashing/flickering stimulation could cause visual fatigue after long-term use [19], [20]. Except for entertainment purposes, Alchalabi et al. [21] have used neurogames to detect attention deficit hyperactivity disorder (ADHD) in patients; the user should control a player to collect some cubics as soon as possible. They used an RBF kernel model to detect these patients and achieved more than 98% accuracy. Liao et al. [22] used a neurogame to measure the attention of users. They extracted focus features, which was the inverse of average power in alpha rhythms; they used this feature to measure the focus level (a bar in the game environment).

mVEPs suggest an elegant BCI paradigm with low luminance values and contrast. A dominant negative peak at the latency of 135–180 ms characterizes these potentials. They have concluded that the negative peak is the motion-specific component [23]. Low luminance and contrast would decrease visual fatigue in long-term use. Guo et al. introduced a new mVEP paradigm with five rectangular white boxes and a black border with a total length of 1.24° and height of 0.76° and a red bar with a length of 0.66°, moving from the left to the right side in 140 ms to elicit mVEP responses in EEG. They used four classification methods to detect the target stimulus, such as P2 area, N2 area, P2+N2 area, and stepwise linear discriminant analysis (SWLDA). The SWLDA classification method obtained higher accuracy than the others. They have also seen that the combination of N2 and P2 gained a higher accuracy than using P2 or N2 lonely. Among all VEPs, mVEP displays the largest amplitudes and the lowest inter- and intrasubject variabilities. It, therefore, seems to be a promising signal for the BCI application [24]. Chen et al. [25] have designed a new multi-target mVEP stimulus to

measure the accuracy of classification between up, down, left, and right; they achieved 72% accuracy averaged in all subjects at a 5.1° visual angle, also they have seen that by increasing the visual angle between these stimuli and their targets, the accuracy of classification would decrease. Zhang et al. proposed a new method to increase the information transfer rate (ITR) and decrease the number of repetitions in mVEP-based BCI systems. They used P1, N2, and P2 amplitudes as a criterion rate to stop the repetitions. The results showed that the practical ITR was improved by using this method [4]. Delon-Martin et al. [26] evaluated four types of mVEP stimulation: translation, contraction, expansion, and rotation, and saw that the amplitude of P1, N2, and P2 components is lower in translation than the others and a new component P180 was discovered in rotation stimulation.

Krusiński et al. [27] used the SWLDA classification method to detect the target character for the first time in the P300 speller. They used three decimation factors to downsample the data, and the results showed that by decimation factor 12, SWLDA achieved higher accuracy in online and offline modes. Riccio et al. [28] used the SWLDA classification method to detect the target character in P300 speller for amyotrophic lateral sclerosis patients. They have achieved 87.5% and 97.5% average accuracy among all subjects in offline and online modes, respectively. Jin et al. [29] proposed a combined BCI based on p300 and mVEPs. They compared this new BCI based on each simple approach. They have seen that the combined BCI outperformed the others in online and offline modes. The accuracy and ITR were 96% and 26.7 bit/min, respectively. They have used the Bayesian linear discriminant and Fisher linear discriminant to classify between target and non-target stimuli. Beveridge et al. [30] designed a computer game based on mVEPs. They used translation stimulus to control this game. Using the linear discriminant analysis (LDA) model, they achieved 76%, 62%, and 35% in 5, 3, and 1 repetition, respectively. The ITR for 5 and 3 repetitions was maintained between 13.4 and 13.9 bit/min, but for 1 repetition, it decreased to 6.6 bit/min.

With all the mentioned facts about mVEPs, they are an excellent choice to control the game. Over the past years, researchers made many neurogames; the lowest percentage was for the mVEP-based neurogames [4]. The mVEP-based neurogames used translation stimulation to elicit mVEPs in brain signals. For mVEP-based neurogames, they used five virtual buttons in a computer game. All of the mVEP-based computer games used translation stimulation, but in this article, we used three other types of stimulation: contraction, expansion, and rotation. The user should gaze at the target stimulus to get a higher score; they achieved higher classification accuracy (LDA without any feature selection method) by increasing the repetitions of the stimuli. In this article, we built an infinite runner computer game with four types of mVEPs: translation, contraction, expansion, and rotation; in each stimulation, we use one type of mentioned mVEPs. We also extracted three types of features: time domain, wavelet coefficients, and time-domain-wavelet coefficients with three classification methods: least absolute shrinkage and selection operator (Lasso) LDA, SWLDA, and spatial-temporal discriminant analysis (STDA), to classify target and non-target

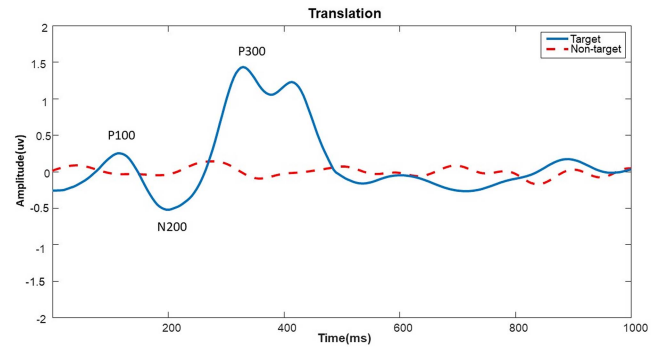


Fig. 1. mVEP smoothed response including P100, N200, and P300 neural potentials for a subject at translation stimulation. Due to the more pronounced peaks, there is a clear difference between the target (blue line) and non-target (red dash line) responses to the moving stimulus.

stimuli. We used nine classification methods to detect the target stimulus at each type of stimulation. We aim to evaluate the possibility of controlling a computer game with rotation, contraction, and expansion stimuli. We also evaluate the game control accuracy in each type of stimulation with all of the nine classification methods in offline and online modes.

II. METHODS

A. Data Acquisition Setup

Eight healthy subjects, aged 22–25 years old, all male, and had no previous experience using BCI systems, volunteered for our experiment. Before the test, we described the task to them and ensured they had no problem performing it.

EEG signals were recorded from the subjects by the 32-channel EEG-Bayamed recording device with a sampling rate of 250 Hz. According to the article presented in [30], to record the visual-evoked potentials resulting from mVEP stimulation, which include three essential components, P100, N200, and P300 (as shown in Fig. 1), 12 channels according to the international 10–20 system of electrode placement covering occipital and parietal areas Cz, TP7, CPz, TP8, P7, P3, Pz, P4, P8, O1., Oz, and O2 were used in the 32 available channels (as shown in Fig. 2). We used the right ear as the reference potential and the left ear as the ground for the device. We used MATLAB software to process the EEG signals.

B. BCI Calibration

The computer game made in this article is known as the infinite runner. We used unity game development software to design this computer game. Unity software [31] is a physics engine that is used to make computer games. Fig. 3(a) shows that five red arrows appear a few seconds before the player hits an obstacle in the game environment. Each of these five arrows has a corresponding virtual button above. As the player approaches these arrows, one of them turns green in a random order, indicating the target stimulus.

The computer game consists of four types of stimuli: translation, contraction, expansion, and rotation. Each type of stimulation contains 25 obstacles, and the number of repetitions of

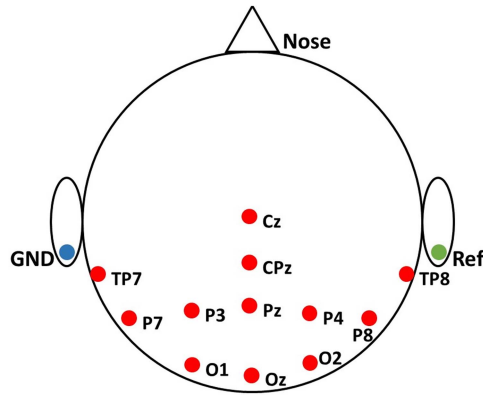


Fig. 2. 12-channel EEG montage used for the study, covering the occipital areas (electrodes colored in red). The right ear was used as a reference (ref), and the left ear electrode (Gnd) was used as a ground.

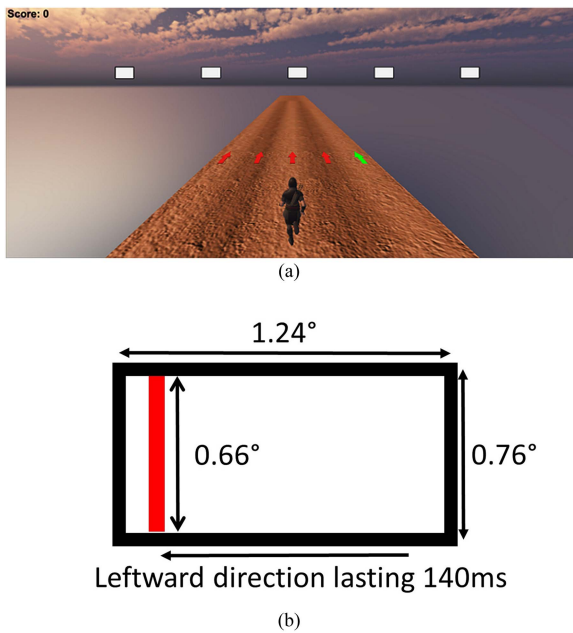


Fig. 3. (a) Computer game environment in the translation stimulation shows the user's score in the upper left corner of the game. (b) Single mVEP translation stimulus as used in this study. The red line inside the black box moves horizontally from right to left lasting for 140 ms.

the stimuli differs for each of the five obstacles. For the first five obstacles, each virtual button is stimulated five times, the second five obstacles four times, the third five obstacles three times, the fourth five obstacles two times, and the last five obstacles one time. As the player crosses every five obstacles, the number of repetitions of stimuli in each button decreases by one. Because of this, the player's speed increases during the game. The user must perform all four types of stimulation once without feedback (calibration mode).

To perform the calibration mode, the player must stare at the target stimulus during the game. The decision time is 16, 13, 10, 7, and 4 s for every first, second, third, fourth, and fifth five obstacles, respectively. The task of the player was to mentally count as the target stimulus was activated (motion occurred).

The sequence of the game's stimulations for the user is chosen randomly to weaken or eliminate the effect of other factors, such as emotional fatigue, in the results. The duration of each type of stimulation is about 20 min, and the subject could rest between each type of stimulation for 5 min. Therefore, the total time of the calibration mode was about 95 min. After each type of stimulation, we will have about 375 epochs, 75 of which will be related to target stimuli, and the rest is to non-target stimuli. All data obtained in the calibration run are used to train a linear model to classify target and non-target stimuli. We used this linear model online to apply feedback to the player.

C. Online Game Paradigm

The online mode is the same as the calibration mode, except that the users can play this mode by feedback. In this game, there are 25 obstacles that the user should detect the target stimulus on time, stare at it, and mentally count the number of times it is activated to earn points and change the lane of the runner, which is under the target stimulus; otherwise, the player will encounter obstacles while passing, no points are awarded, and the lane of the runner will not change.

At the end of the calibration mode, the user can rest for one hour before the online mode. The user also does the online mode on the same day due to adaptability. The order of the online mode's stimulations is the same as the calibration mode. The time interval between each type of stimulation and the time duration of each type of stimulation is the same as the calibration mode. As the number of repetitions of stimuli decreases, the score for each obstacle increases, so there are 10, 15, 20, 30, and 45 points for every first, second, third, fourth, and fifth five obstacles. Therefore, the maximum score that users can achieve in each type of stimulation in online mode is 600 points.

D. Translation Stimulus

In the translation stimulation, five rectangles are placed in the game environment, according to Fig. 3(a). The stimulation method in each rectangle will be such that a red vertical bar with a 0.66° visual angle appeared on the right side of the vacant rectangle and moved leftward in 140 ms before it disappeared (motion offset) [as shown in Fig. 3(b)]. A trial is defined as a complete series of motion stimuli of all five red vertical lines across each virtual button.

Each trial is subdivided into five stimulus periods dedicated to the five buttons, respectively, with no overlap. The five brief motion stimuli (140 ms) occur in a random order, so all five stimuli appear before any is repeated. Each of the five stimuli inside a trial also comprises an epoch. An interval of 600 ms is considered between the two trials to prevent overlapping mVEPs.

The translation stimulus consists of a rectangle with a white rectangle and a black border around it. The dimensions of the rectangle are shown in Fig. 3(b). The distance between each stimulus (rectangle) on the monitor screen, according to the article presented in [24], is 4.64° .

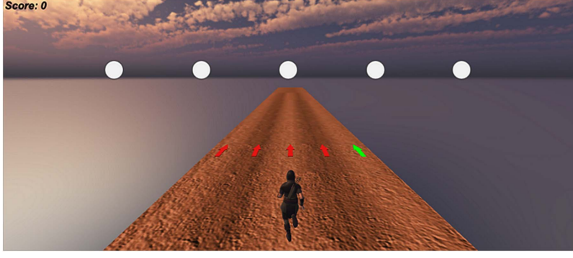


Fig. 4. Computer game environment in the contraction, expansion, and rotation stimulations; the score is shown to the user in the upper left corner of the game.

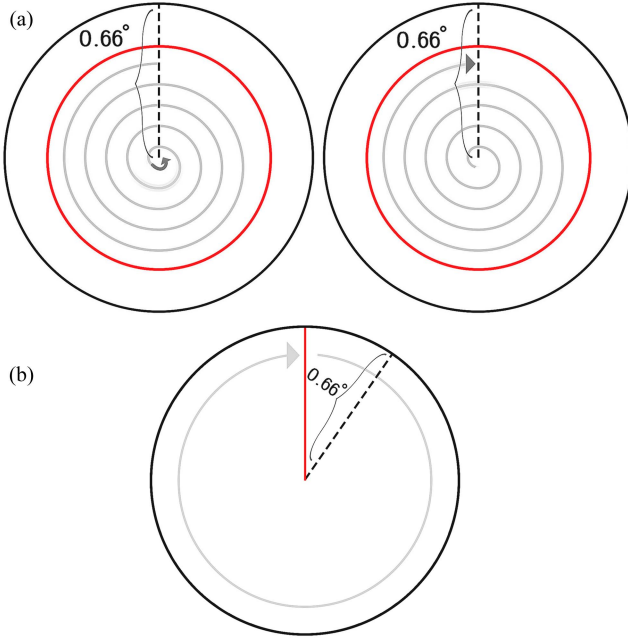


Fig. 5. (a) Contraction stimulus (left picture), the red circle appears on the edge of the white circle and starts to get smaller and disappears after 140 ms. Expansion stimulus (right picture), the red circle's expansion stimulus starts to get bigger and become the same size as the white circle and disappears after 140 ms. (b) Rotation stimulus, the red bar starts to rotate 360° clockwise and disappears after 140 ms.

E. Contraction, Expansion, and Rotation Stimuli

We have five white circles (virtual buttons) with a black border around them in the game environment for contraction, expansion, and rotation stimulations, according to Fig. 4. In the stimulations mentioned earlier, the five stimuli occur in each white circle in random order (so that all five stimuli appear before any is repeated).

For contraction, a hollow red circle appears on the edge of each virtual button in the game environment, and it gets smaller to become a dot and disappears in 140 ms [as shown in Fig. 5(a)].

For expansion, a hollow red circle appears inside each virtual button in the game environment as a dot, and it gets bigger (until it becomes the same size as the white circle) and disappears in 140 ms [as shown in Fig. 5(a)].

For rotation, it is a red bar of the same radius as the white circles, which appears to rotate 360° and disappears in 140 ms [as shown in Fig. 5(b)].

In this article, the user distance from the monitor is 50 cm. The distance between each stimulus (circle) on the monitor screen, according to the article presented in [24], is 4.64°.

F. Data Preprocessing and Feature Extraction

A bandpass Butterworth filter with order 5 filtered the data digitally in the frequency range of 0.5–25 Hz for preprocessing. We extracted target and non-target epochs, which lasted for 1 s (250 samples) from the beginning of the motion onset. Because the maximum available frequency in each epoch was 25 Hz, the new sampling rate must be at least twice the maximum frequency of the signal. Data epochs were resampled at 50 samples/s to prevent aliasing.

Extracting time sample features: After resampling the data, the time samples in each epoch are changed from 250 to 50. Therefore, the feature vector for each epoch has 50×12 (number of channels) = 600 features.

Extracting wavelet features: After changing the sampling rate, using the mother wavelet function db4 [32] with level decomposition = 5, approximation and detail values for each epoch in each channel were calculated and used as a feature vector, which has 633 features.

Extracting time samples–wavelet features: We combined both feature vectors (temporal and wavelet feature vectors) to increase the classification accuracy. A feature vector with 1233 features was built for each epoch.

G. Classification Methods

Lasso–LDA: The method (least absolute shrinkage and selection operator), or Lasso for short, selects some properties and sets the value of others to zero. This method is suitable for classification and regression and tries to select a subset of features.

Assuming that our data are (X^i, y_i) , $i = 1, 2, 3, 4, \dots, N$, where $X^i = (X_{i1}, \dots, X_{ip})^T$ is the matrix of our prediction variables (matrix for feature vectors), y_i is the matrix for our responses (label matrix), and N is the number of observations. We assume that the observations are statistically independent of each other and that the matrix of the X_{ij} prediction variable is normalized, which means that the observations have a mean of zero and a variance of one relative to each other. According to the matrix $\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_p)^T$, the Lasso method calculates the variables $(\hat{\alpha}, \hat{\beta})$ according to the following equation:

$$(\hat{\alpha}, \hat{\beta}) = \arg \min \left\{ \sum_{i=1}^N \left(y_i - \alpha - \sum_j \beta_j x_{ij} \right)^2 \right\} \quad \text{subject to } \sum_j |\beta_j| < t. \quad (1)$$

The above equation is $t \geq 0$ and is a regulatory parameter. The parameter $t \geq 0$ controls the amount of shrinkage calculated for the estimate. Suppose the parameter $\hat{\beta}_j^o$ is the least squares of the estimation $t_0 = \sum |\beta_j^o|$. The values $t < t_0$ shrinkage takes the answers to zero, so some coefficients will be precisely equal to zero. By applying this method, based on the least squares of

error, the features whose coefficients are opposite to zero are selected and given to the LDA classifier for classification [33].

SWLDA: Stepwise linear discriminant analysis [34], or SWLDA for short, is a method for selecting appropriate features in a multivariate regression model using the following equation:

$$\omega * X - b = 0. \quad (2)$$

In this method, forward and backward-step regression is performed. Based on the article presented in [27] to detect the evoked potentials of P300, assuming the start without any initial conditions for the model, the statistically significant features with p -values < 0.1 are added to the model. After adding any new feature to the model, it performs a backward-step linear regression to remove features that are not statistically significant and have p -values > 0.15 .

According to the article presented in [28], the above steps are repeated until 60 statistically significant features are selected.

STDA: The method (spatial-temporal discriminant analysis), or STDA for short, is a method that maximizes the classification information between two target and non-target classes by finding two matrices in spatial and temporal dimensions in a participatory way. This method significantly reduces the dimensions of the features [35].

To explain the STDA method, we first consider the following two operators:

$$X \times_1 W_1^T = W_1^T X \quad (3)$$

$$X \times_2 W_2^T = X W_2. \quad (4)$$

In the above equations, the matrix $X \in R^{D1 \times D2}$, the matrix of our properties in space-time, and the matrices $W_1 \in R^{D1 \times L}$ and $W_2 \in R^{D2 \times L}$ are also the projection matrices for space and time, respectively. It is also assumed that the dimension value L is smaller than the dimensions $D1$ and $D2$ ($L^2 \ll D1 D2$).

The STDA method performs the optimization operation alternately in space and time for each sample in the X matrix. For the k optimization ($k = 1, 2$), we first perform the projection operation for each of the spatiotemporal samples of the matrix $X_i \in R^{M \times P}$ in 3- k mode

$$\begin{cases} Y_i^1 = X_i \times_2 W_2^T \text{ for } k = 1 \\ Y_i^2 = (X_i \times_1 W_1^T)^T \text{ for } k = 2. \end{cases} \quad (5)$$

Then, the scattering matrix between classes S_B^K and the scattering matrix within the classes can be calculated from (6) and (7) through the spatiotemporal samples, which are illustrated by the above equation

$$S_B^K = \sum_{c=1}^2 N_c (\bar{Y}_c^k - \bar{Y}^k) (\bar{Y}_c^k - \bar{Y}^k)^T \quad (6)$$

$$S_W^K = \sum_{c=1}^2 \sum_{i \in \tau_c} (Y_{c,i}^k - \bar{Y}_c^k) (Y_{c,i}^k - \bar{Y}_c^k)^T. \quad (7)$$

In the above equations, the matrices $Y_{c,i}^k$ are the first sample shown in class c , and $\bar{Y}_c^k$ and $\bar{Y}^k$ are the average of the samples shown in class c and the average of all the samples shown,

respectively. According to the linear classification algorithm, the k th of the projection matrix is calculated by the following equation:

$$\tilde{W}_k = \arg \max_{w_k} \left(\frac{\text{tr}(W_k^T S_B^K W_k)}{\text{tr}(W_k^T S_W^K W_k)} \right). \quad (8)$$

The above equation can be solved by the generalized problem method of eigenvalues (9)

$$S_B^K W_k = S_W^K W_k \wedge_k. \quad (9)$$

In the above equation, the eigenvalues of the matrix W_k correspond to the largest value of L . The eigenvalues in the matrix $\wedge_k$ are used to form a projection matrix $\tilde{W}_k$. The iteration process is used to optimize the projection matrices in the dimensions of space and time.

After performing the optimization operation, the projection matrices $\tilde{W}_1 \in R^{M \times L}$ and $\tilde{W}_2 \in R^{P \times L}$ are obtained. Each of the $X_i \in R^{M \times P}$ specimens are projected to a new space $f_i \in R^{L^2}$ ($L^2 \ll MP$) by the estimated projection matrices (10)

$$f_i = \text{vec} \left(X_i \times_1 \tilde{W}_1^T \times_2 \tilde{W}_2^T \right), i = 1, 2, \dots, N. \quad (10)$$

In the above equation, $\text{vec}(\cdot)$ represents the vectorization operator. According to the illustrated samples f_i and the corresponding labels $l_i \in \{1, 2\}$ ($i = 1, 2, \dots, N$), a projection vector $w_f \in R^{L^2}$ according to formula (11) by a simple LDA will be separable. Assuming the evaluation data $\hat{X} \in R^{M \times P}$, and then representing them by the projection matrices and forming the vectors $\hat{f} \in R^{L^2}$ from the matrix $\hat{X}$ scores can be obtained, which can be calculated by a linear separator calculated from the following equation [35]:

$$H = W_f^T \hat{f}. \quad (11)$$

To compare the accuracies of different methods, the two-sample t -test was used to calculate p -values.

H. Information Transfer Rate

Wolpaw et al. [1] suggested evaluating BCI performance using *bit rate*. Bit rate, or *ITR*, is the standard method for measuring communication and control systems. This measurement incorporates both speed and accuracy in a single value. Bit rate has been most commonly applied to assess the performance of an entire BCI system [36]. ITR, reported in bits/min, is calculated in (12), where N is the number of classes and P is the probability of correct classification [30]

$$B \left(\frac{\text{Trials}}{\text{min}} \right) = \left(\log_2(N) + P \log_2(P) + (1 - P) \log_2 \left(\frac{1 - P}{N - 1} \right) \right). \quad (12)$$

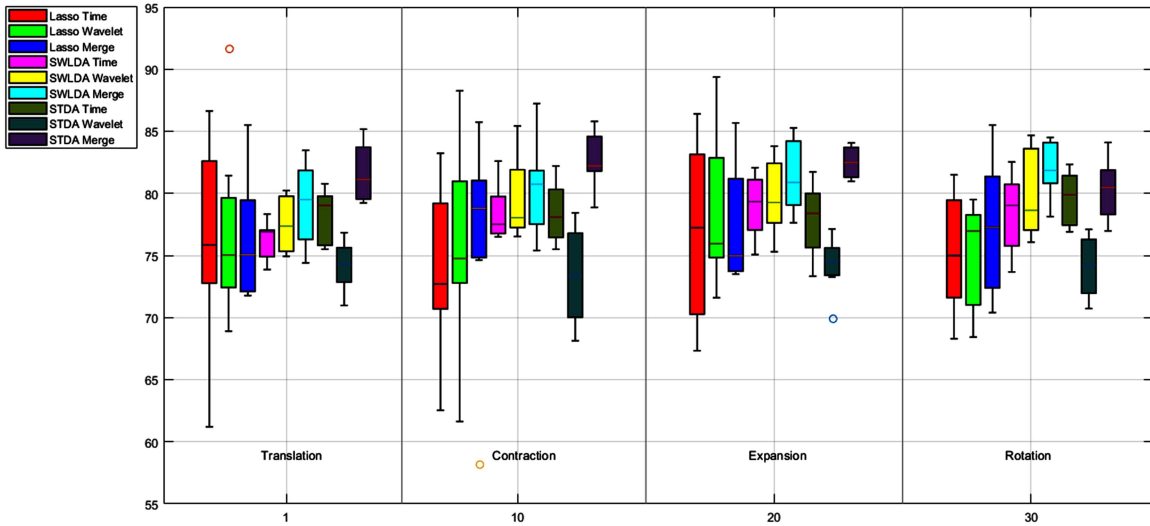


Fig. 6. Box plot diagram of the classification accuracy across subjects for all nine methods at all types of stimulations. From left to right, it indicates the Lasso, SWLDA, and STDA methods with all three types of extracted features. The classification methods were evaluated by five-fold cross validation with 20 repetitions.

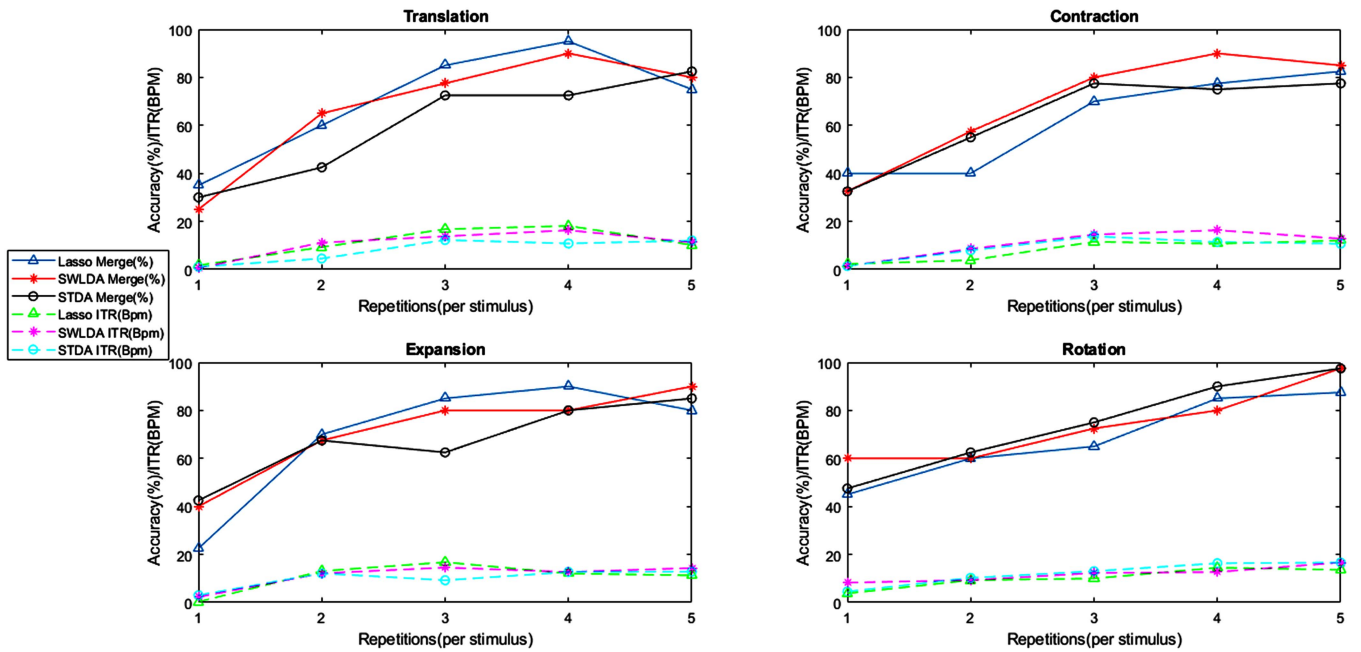


Fig. 7. Diagram of online game accuracy and ITR across all types of stimulations with combination of features by all classification methods; horizontal axis shows the number of trial repetitions and vertical axis shows the values of accuracy(%) and ITR(bits/min).

III. RESULTS

A. Calibration and Online Modes

The box plot diagram (see Fig. 6) demonstrates classification accuracy (target versus non-target) for all nine methods at translation, contraction, expansion, and rotation stimulations from left to right.

The classification accuracy with the Lasso-LDA method using all extracted features has many intersubject variabilities at all stimulations. For the SWLDA classification method, the accuracy has fewer intersubject variabilities than the Lasso, and the highest accuracy is achieved by the time-domain-wavelet

coefficients as features (i.e., SWLDA Merge). SWLDA Merge significantly outperformed all other methods in rotation stimulus ($p < 0.05$, based on two-sample t -tests).

The STDA classification method results demonstrate that the classification accuracy has a few intersubject variabilities, and the highest accuracy is for the time-domain-wavelet coefficients as features (i.e., SWLDA Merge). STDA Merge significantly outperformed all other methods in translation, contraction, and expansion stimuli ($p < 0.05$, based on two-sample t -tests).

Fig. 7 represents the average classification accuracy across all subjects in translation, contraction, expansion, and rotation

TABLE I

TOTAL SCORES OBTAINED BY THE SUBJECTS IN THE ONLINE MODE FOR ALL TYPES OF STIMULATIONS USING THE TIME SAMPLES-WAVELET COEFFICIENTS AS FEATURES AND THE LASSO METHOD

	Translation	Contraction	Expansion	Rotation
Sub1	300	475	350	350
Sub2	205	210	310	385
Sub3	290	280	320	410
Sub4	545	320	320	120
Sub5	355	350	330	405
Sub6	480	420	420	295
Sub7	305	395	345	525
Sub8	390	345	345	420
Total	2870	2795	2740	2910

TABLE II

TOTAL SCORES OBTAINED BY THE SUBJECTS IN THE ONLINE MODE FOR ALL TYPES OF STIMULATIONS USING THE TIME SAMPLES-WAVELET COEFFICIENTS AS FEATURES AND THE SWLDA METHOD

	Translation	Contraction	Expansion	Rotation
Sub1	235	210	310	345
Sub2	275	405	140	410
Sub3	380	455	430	515
Sub4	210	275	355	445
Sub5	385	340	355	420
Sub6	420	415	420	275
Sub7	310	390	585	405
Sub8	360	305	415	435
Total	2575	2795	3010	3250

stimulations. Time-domain-wavelet coefficients were used as features in online mode. The blue, red, and black lines show Lasso/LDA, SWLDA, and STDA classification methods. The horizontal axis indicates the number of stimulation repetitions in each virtual button, and the vertical axis indicates the classification accuracy. As the number of repetitions of stimulation decreases, the classification accuracy in the online mode decreases too.

The ITR for each number of repetitions was also calculated according to (12). The results are shown in Fig. 7 for each type of stimulation at the same plot of the accuracy results.

B. Game Scores Obtained by Subjects in Online Mode

Tables I–III present the total scores obtained by the subjects in the online mode by using the time-domain-wavelet coefficients as features and the Lasso/LDA, SWLDA, and STDA methods for classification at all types of stimulations.

Results show that the highest scores obtained by the subjects are at rotation stimulation for all classification methods.

C. Grand Averages and Topographical Maps

The grand average responses and the topographical maps when the P300 component occurs at four different types of stimulation are shown in Fig. 8. The three high-ranked electrodes

TABLE III

TOTAL SCORES OBTAINED BY THE SUBJECTS IN THE ONLINE MODE FOR ALL TYPES OF STIMULATIONS USING THE TIME SAMPLES-WAVELET COEFFICIENTS AS FEATURES AND THE STDA METHOD

	Translation	Contraction	Expansion	Rotation
Sub1	285	350	260	425
Sub2	265	260	375	355
Sub3	355	345	210	330
Sub4	150	190	465	265
Sub5	415	435	455	340
Sub6	285	300	375	320
Sub7	285	370	480	555
Sub8	355	375	275	450
Total	2395	2625	2895	3040

are Cz, CPz, and Pz when the N200 and P300 occur. It could be seen from the grand average responses that the P300 components in the translation and expansion stimulations are higher than the contraction and rotation. On the contrary, the N200 components in the contraction and rotation stimulations are higher than the translation and expansion.

IV. DISCUSSION

A. Translation Stimulus

In this stimulus at calibration mode, the STDA method [35] with the combination of features has the best accuracy. The worst accuracy is classification by the STDA method with wavelet features. The Lasso/LDA method's [33] accuracy has many intersubject variabilities. The least intersubject variabilities are for the SWLDA method [27], [28], [34] with time-domain features. The accuracy improves when the combination of features is used in the SWLDA method. One of the subjects obtained more than 90% accuracy when using the Lasso/LDA method with wavelet features.

In the online mode, we used a combination of features for classification. The Lasso/LDA method outperformed the others at some repetitions; when the repetitions increased, the accuracy of the STDA and SWLDA methods became better.

The ITR was better for this stimulation type using the Lasso-LDA classification method. It achieved the highest ITR [30] in four repetition modes at the Lasso/LDA method.

B. Contraction Stimulus

In calibration mode, the best accuracy is achieved by the STDA method with the combination of features. The worst classification accuracy is for the STDA method with wavelet features. The accuracy did not improve when the combination of features was used with the SWLDA method. The intersubject variabilities at SWLDA and Lasso/LDA methods are like translation stimulus, except that the classification accuracy improves when the features combine at Lasso/LDA method. The classification accuracy with the STDA method is higher than the other stimulations.

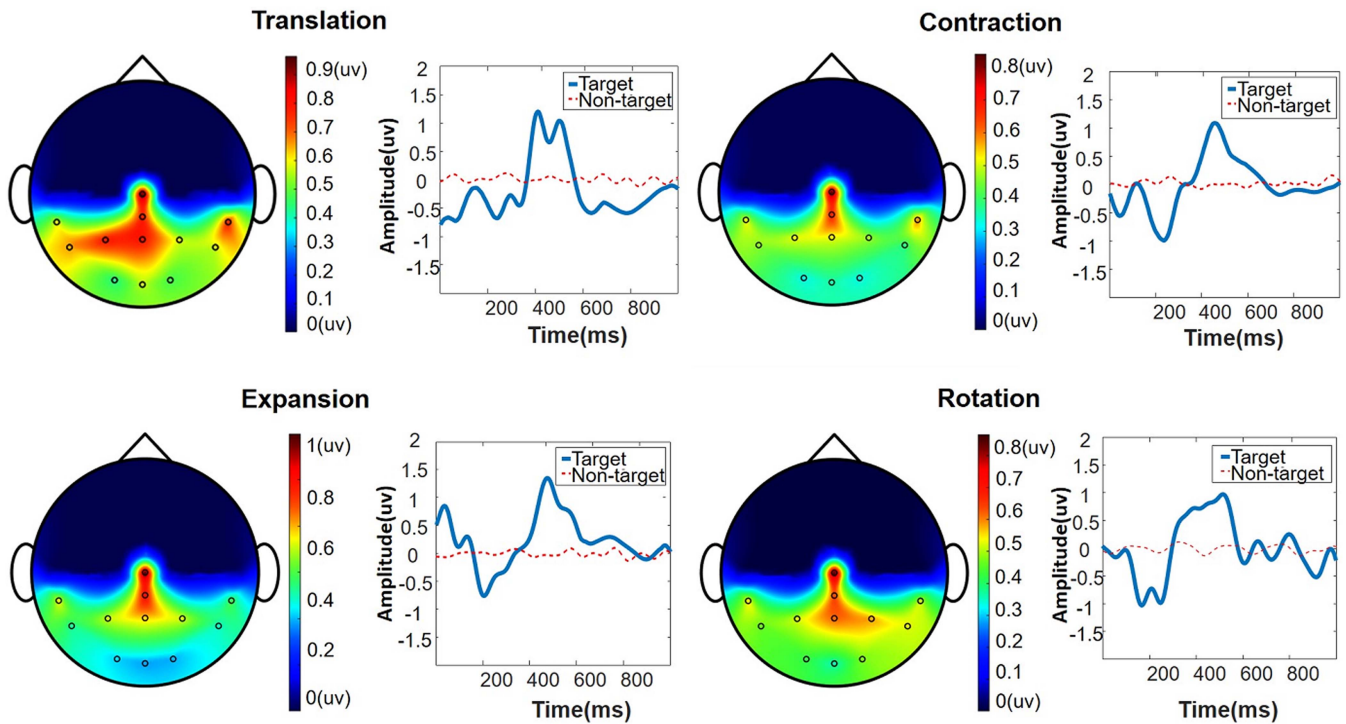


Fig. 8. Grand average responses and the topographical maps for all types of stimulations. The topographical maps are plotted based on the responses on all 12 recorded EEG channels, including Cz, TP7, CPz, TP8, P7, P3, Pz, P4, P8, O1, Oz, and O2 when the P300 component occurs.

The classification results in the online mode demonstrate that the SWLDA method outperforms the others. The accuracy at the one-repetition mode with Lasso/LDA method is higher than the other methods, but when the repetition of the stimulus increases, the accuracy of this method decreases.

The ITR was better for this type of stimulus at the SWLDA classification method than the other methods. The mean ITR for this stimulation was not good compared with the other stimulations at Lasso/LDA and SWLDA classification methods.

C. Expansion Stimulus

The results in calibration mode show that the best classification accuracy is for the SWLDA method when the combination of features is used. The worst accuracy is for the STDA method with wavelet features, and the least intersubject variabilities are with the combination of features.

In online mode, the classification accuracy is higher with Lasso/LDA model at some repetitions. The STDA method is better in one-repetition mode, but the SWLDA method is better than the others in the five-repetition mode.

The ITR was better for this type of stimulation at the SWLDA classification method than the other methods.

D. Rotation Stimulus

The SWLDA method has the best accuracy and the least intersubject variabilities in the calibration mode. The Lasso/LDA method has many intersubject variabilities, and the accuracy is improved when the combination of features is used.

In online mode, the classification accuracy of SWLDA and STDA methods is strictly ascending. We can see that the accuracy increases by increasing the repetitions of stimulation. SWLDA and STDA methods obtained more than 95% classification accuracy in the five-repetition mode. The SWLDA method obtained 60% accuracy in one-repetition mode, which is better than the other stimulations in this mode. The STDA method is better at the other repetitions mode. We conclude that the subjects got higher accuracy and scores at the rotation stimulus than at the other stimulations.

This type of stimulation achieved the highest ITR at five-repetition modes. As we can see, the mean of its ITR at all methods except Lasso-LDA is higher than the other types of stimulations.

E. BCIs and Computer Games

People with movement disorders who cannot use conventional tools, such as a mouse, keyboard, or other game controllers, can benefit from BCI-based computer games to control a computer game [2], [3]. Direct conversion of electrophysiological signals of the brain into computer commands will allow them to do this [1]. Today, computer games based on BCI are used to rehabilitate stroke patients. In addition, healthy people can use these types of games for recreational purposes [4]. Computer games based on BCI can be used to diagnose some mental diseases, such as ADHD [21].

mVEP stimuli can be incorporated into the game environment due to their high sensitivity. These stimuli cause less long-term

visual fatigue for people due to their lower contrast and brightness than P300 and SSVEP stimuli that are flashed. For this reason, it is more appropriate to use them in a computer game than other stimulations [30].

It is important to note that the use of the mVEP stimuli in a BCI-based game has more limitations than the traditional interactive games (e.g., via keyboard or mouse). In fact, there is a cost-benefit tradeoff between increasing efficiency and the number of repetitions of stimuli. By increasing the number of repetitions of stimuli, the efficiency of BCI increases, but the time required to generate the output of BCI also increases. For example, it takes more time to generate an output with one to five stimulus repetitions from 4 to 16 s in the current BCI-based game than in the traditional interactive mode. At the same time, it should also be noted that this limitation exists for other types of BCI-based games, more or less due to the addition of an interface system in the output production.

In the previous studies where mVEPs were used in computer games, only the translation stimulus was used in the computer game environment for the controller. They have used the LDA classification model to recognize the target stimulus [4], [17], [18]. In this research, we have used four different types of stimulation, three classification methods, and three different feature extraction methods to detect the target stimulus. Our goal in this project was to achieve the highest accuracy in controlling a computer game in different types of stimulation by these nine combined methods. Especially in the online mode, our goal was to be able to identify the best type of mVEP stimulation with the best classification method in detecting the target stimulus to increase the accuracy of controlling a computer game by these types of stimuli.

V. CONCLUSION

This study investigated the feasibility of controlling a computer game by contraction, expansion, and rotation of mVEP stimuli in addition to the translation. The performance of these four types of stimuli in controlling a computer game was also compared. For the classification of three different features, including time domain, wavelet coefficients, and time-domain-wavelet coefficients, classification methods, including Lasso, SWLDA, and STDA, and a linear classifier were used to classify target versus non-target stimuli. In this study, all of these methods were compared for all four types of stimulations. Another purpose of this study was to compare stimuli by applying the mentioned methods to determine the best classification method and mVEP stimulus to build computer games or BCI systems based on these potentials.

In the Lasso/LDA method for the translation and expansion stimuli, this classification method works better in the four-repetition mode than in the five-repetition mode because the user is trained to detect the target stimulus on time.

In addition to the translation stimulus, three other mVEP stimuli, including contraction, expansion, and rotation, were used to control a computer game. Previous mVEP-based BCI games have only used translation stimulus to control the game. In addition to the time-domain features, this study also used wavelet coefficients and the combination

of time-domain and wavelet coefficients. This work also employed three classification methods, Lasso, SWLDA, and STDA, which have not been used previously to detect the target stimulus in mVEP-based computer games. We have observed that the best classification accuracy and the ITR are obtained using time-domain-wavelet coefficients as features with SWLDA/STDA classification methods using the rotation stimulus. We suggest the rotation stimulus as a good choice to design an mVEP-based BCI system or a neurogame.

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